AI-Powered Customer Retention: Hybrid Churn Prediction with BERT-Driven Narratives

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Problem Statement and Motivation

The Gap: Standard ML models often act as "black boxes," providing binary "Churn" or "No Churn" labels without explaining the underlying customer sentiment or context.

The Goal: Build a hybrid predictive system that combines the statistical power of Random Forest with the semantic depth of BERT to generate human-readable customer narratives.

Impact: Empowers retention managers to move beyond raw numbers, providing actionable insights into "Frustrated" or "At-Risk" customers through an interactive natural language interface.

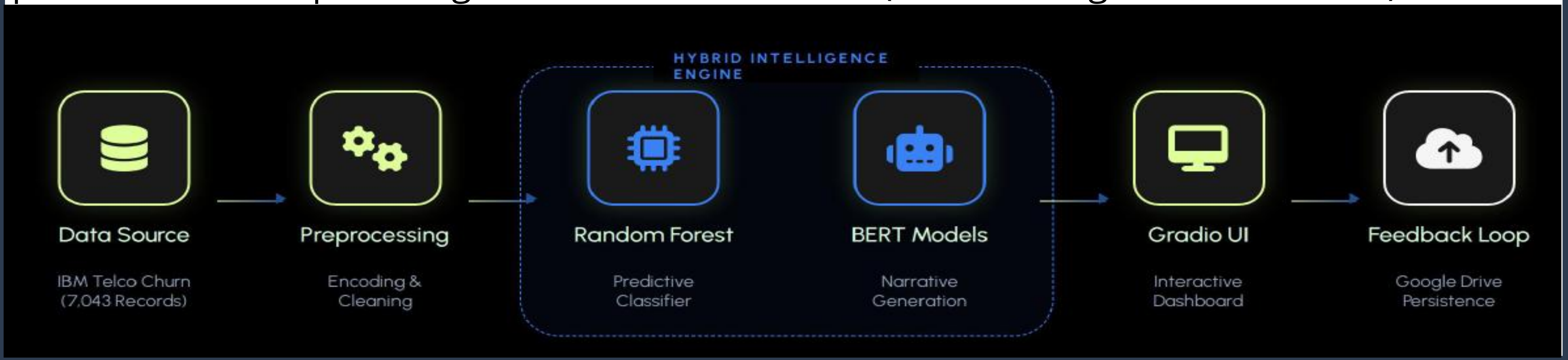
Methodology and System Architecture

Pipeline Logic: A four-phase workflow comprising Data Ingestion, Preprocessing, Supervised ML, and Generative AI Augmentation.

Modeling: Trained a Random Forest ensemble on 31 engineered features to establish a robust predictive baseline for binary churn classification.

AI Augmentation: Utilized BERT (bert-base-uncased) to transform structured account data into 768-dimensional semantic narratives for better interpretability.

User Interaction (UI): Interactive Gradio dashboard that translates model probabilities into plain-English "Risk Assessments" (Neutral, Negative, Frustrated).



Conclusion and Future Work

Key Finding: Proved that a Hybrid Pipeline (RF + BERT) provides significantly deeper customer context than standalone binary classifiers.

Unified Insights: Successfully bridged the gap between structured billing data and semantic narratives for a "complete" view of at-risk customers.

Future Roadmap:
Optimization: Implement SMOTE oversampling and hyperparameter tuning to improve minority class recall.

Deep Learning: Transition to XGBoost and fine-tuned domain-specific BERT models for higher predictive precision.

Dataset and Knowledge Base

Structured Data: IBM Telco Customer Churn dataset. Includes 7,043 customer records across 21 initial demographic and service attributes.

Semantic Knowledge Base: Integrated BERT (bert-base-uncased) to generate 768-dimensional semantic embeddings from customer narratives.

Processing: Excluded customerID, handled missing values via errors='coerce' for TotalCharges, and applied One-Hot/Binary encoding to expand features to 31 dimensions.

Results (Evaluation and Performance)

Model Accuracy: Achieved a robust 78.75% overall accuracy using a Random Forest classifier.

Class Performance: High performance on non-churners (0.86 F1-score); recall of 0.48 on churners due to class imbalance.

Interpretability: BERT-based augmentation successfully converted into semantic narratives, enabling human-readable sentiment profiling (Neutral, Negative, Frustrated).

Operational Utility: The Gradio dashboard enabled real-time inference, bridging the gap between ML outputs and business users.

Demo and Application